Entropy and Gibbs Free Energy

Answer key with worked solutions

A. Understanding

1. Concepts.

> (a) A measure of how **spread out** energy and matter are (the number of ways particles and energy can be arranged). (b) In a spontaneous process, the **total entropy of the universe increases**. (c) ΔG = ΔH − TΔS: melting has a positive ΔS, and at 25 °C the **TΔS term is larger than ΔH**, so ΔG is negative even though ΔH is positive.

2. Signs of ΔS.

> (a) **+** (liquid → gas) (b) **−** (3 mol gas → 2 mol) (c) **+** (ordered solid → mobile ions) (d) **+** (a gas is formed) (e) **close to zero** (2 mol gas on each side)

3. Four cases.

> (a) **All temperatures** (ΔH −, ΔS +) (b) **Never** (ΔH +, ΔS −) (c) **Only at low T**, below $T = \dfrac{60\ \text{kJ}}{0.150\ \text{kJ/K}} = 400\ \text{K}$ (d) **Only at high T**, above $T = \dfrac{90\ \text{kJ}}{0.200\ \text{kJ/K}} = 450\ \text{K}$

B. Calculations

4. ΔS° of methane combustion.

> $\Delta S^\circ = [213.8 + 2(70.0)] - [186.3 + 2(205.2)] = 353.8 - 596.7 =$ **−242.9 J/K**
>
> Negative because **3 mol of gas** become **1 mol of gas** (the water is liquid).

5. ΔH° and ΔG°.

> $\Delta H^\circ = [-393.5 + 2(-285.8)] - [-74.6 + 2(0)] =$ **−890.5 kJ**
>
> $\Delta G^\circ = -890.5\ \text{kJ} - (298.15\ \text{K})(-0.2429\ \text{kJ/K}) = -890.5\ \text{kJ} + 72.42\ \text{kJ} =$ **−818.1 kJ** (spontaneous)

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6. Melting ice.

> (a) $T = \dfrac{6.01\ \text{kJ/mol}}{0.0220\ \text{kJ mol}^{-1}\,\text{K}^{-1}} = \mathbf{273\ K}$ (0 °C, the melting point)
> (b) At 263.15 K: $\Delta G = 6.01\ \text{kJ/mol} - (263.15\ \text{K})(0.0220\ \text{kJ mol}^{-1}\,\text{K}^{-1}) = \mathbf{+0.221\ kJ/mol}$: ice does **not** melt.
> At 283.15 K: $\Delta G = 6.01\ \text{kJ/mol} - (283.15\ \text{K})(0.0220\ \text{kJ mol}^{-1}\,\text{K}^{-1}) = \mathbf{-0.219\ kJ/mol}$: ice **melts** spontaneously.

7. Limestone at 1200 K.

> $\Delta G^\circ = 179.2\ \text{kJ} - (1200\ \text{K})(0.1602\ \text{kJ/K}) = 179.2\ \text{kJ} - 192.2\ \text{kJ} = \mathbf{-13.0\ kJ}$: spontaneous (1200 K is above the switching temperature, 1119 K).

8. K from ΔG°.

> $\ln K = -\dfrac{\Delta G^\circ}{RT} = -\dfrac{1.00 \times 10^4\ \text{J/mol}}{(8.314\ \text{J mol}^{-1}\,\text{K}^{-1})(298.15\ \text{K})} = -4.034$
> $K = e^{-4.034} = \mathbf{0.0177}$: K is less than 1, so **reactants** are favoured.

C. Challenge

9. Daniell cell.

> (a) $\Delta G^\circ = -nFE^\circ = -(2\ \text{mol})(96\,485\ \text{C/mol})(1.10\ \text{V}) = -2.123 \times 10^5\ \text{J} = \mathbf{-212\ kJ}$ (C × V = J)
> (b) $\ln K = -\dfrac{\Delta G^\circ}{RT} = \dfrac{2.123 \times 10^5\ \text{J/mol}}{(8.314\ \text{J mol}^{-1}\,\text{K}^{-1})(298.15\ \text{K})} = 85.63$, so $K = e^{85.63} \approx \mathbf{1.5 \times 10^{37}}$
> The reaction goes essentially **to completion**: at equilibrium almost no Cu^{2+} would remain.

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